

```

%% Given Data
R = 3;
x = [3*R 5*R 6*R 8*R 7*R 4*R 2*R 9*R];
N = 8;

% Bit-Reversal Reordering
x_br = [x(1) x(5) x(3)
        x(7) x(2) x(6) x(4) x(8)];

%% Twiddle Factors
W8_0 = exp(-1j*2*pi*0/8);
W8_1 = exp(-1j*2*pi*1/8);
W8_2 = exp(-1j*2*pi*2/8);
W8_3 = exp(-1j*2*pi*3/8);

%% Stage 1 (2-point DFTs)
S1 = zeros(1,8);
for k = 1:2:8
    S1(k) = x_br(k) + x_br(k+1);
    S1(k+1) = x_br(k) - x_br(k+1);
end

%% Stage 2 (4-point DFTs)
S2 = zeros(1,8);

S2(1) = S1(1) + S1(3);
S2(3) = S1(1) - S1(3);
S2(2) = S1(2) + W8_2*S1(4);
S2(4) = S1(2) - W8_2*S1(4);

%% Stage 3 (8-point DFT)
X = zeros(1,8);

X(1) = S2(1) + S2(5);
X(5) = S2(1) - S2(5);

S2(5) = S1(5) + S1(7);
S2(7) = S1(5) - S1(7);
S2(6) = S1(6) + W8_2*S1(8);
S2(8) = S1(6) - W8_2*S1(8);

X(2) = S2(2) + W8_1*S2(6);
X(6) = S2(2) - W8_1*S2(6);

X(3) = S2(3) + W8_2*S2(7);
X(7) = S2(3) - W8_2*S2(7);

X(4) = S2(4) + W8_3*S2(8);
X(8) = S2(4) - W8_3*S2(8);

%% Display Result
disp('8-point DFT X[k] using DIT-FFT:');
disp(X);

%% Verification (Optional)
disp('Verification using MATLAB');
fft(:)');
disp(fft(x));

clc; clear; close all;

%% Given data
R = 3;
x = [3*R 5*R 6*R 8*R 7*R 4*R 2*R 9*R];
N = 8;

%% Twiddle factors
W8_0 = exp(-1j*2*pi*0/8);
W8_1 = exp(-1j*2*pi*1/8);
W8_2 = exp(-1j*2*pi*2/8);
W8_3 = exp(-1j*2*pi*3/8);

%% ----- Stage 1 (8-point DIF)
-----
S1 = zeros(1,8);
S1(1) = x(1) + x(5);
S1(5) = (x(1) - x(5))*W8_0;
S1(2) = x(2) + x(6);
S1(6) = (x(2) - x(6))*W8_1;
S1(3) = x(3) + x(7);
S1(7) = (x(3) - x(7))*W8_2;
S1(4) = x(4) + x(8);
S1(8) = (x(4) - x(8))*W8_3;

%% ----- Stage 2 (4-point DIF)
-----
S2 = zeros(1,8);
% First 4-point group
S2(1) = S1(1) + S1(3);
S2(3) = (S1(1) - S1(3))*W8_0;
S2(2) = S1(2) + S1(4);
S2(4) = (S1(2) - S1(4))*W8_2;

% Second 4-point group
S2(5) = S1(5) + S1(7);
S2(7) = (S1(5) - S1(7))*W8_0;
S2(6) = S1(6) + S1(8);
S2(8) = (S1(6) - S1(8))*W8_2;

%% ----- Stage 3 (2-point DIF)
-----
X_temp = zeros(1,8);
X_temp(1) = S2(1) + S2(2);
X_temp(2) = S2(1) - S2(2);
X_temp(3) = S2(3) + S2(4);
X_temp(4) = S2(3) - S2(4);
X_temp(5) = S2(5) + S2(6);
X_temp(6) = S2(5) - S2(6);
X_temp(7) = S2(7) + S2(8);
X_temp(8) = S2(7) - S2(8);

%% ----- Bit-reversal for output
-----
X = [X_temp(1) X_temp(5) X_temp(3)
     X_temp(7) ...
     X_temp(2) X_temp(6) X_temp(4)
     X_temp(8)];

%% Display result
disp('8-point DFT X[k] using DIF-FFT:');
disp(X);

%% Verification
disp('Verification using MATLAB');
fft(:)');
disp(fft(x));

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